Solutions

Problem Set: The Interpretation of Propositional Formulas
Goldrei §2.3, with emphasis on Theorem 2.2

How to read these. Throughout, "Theorem 2.2" refers to the existence-and-uniqueness of the extension $\overline{v}$; when a step uses only one half I say which. "Relevance" means Problem 8(a). "Unique readability" is the §2.2 fact that every formula of length ≥ 1 has a unique principal connective and hence a unique immediate decomposition. I write $\mathrm{vars}(\varphi)$ for the set of propositional variables occurring in φ, and for a truth assignment v I use the truth-table operations on values freely (e.g. $v(\neg\theta) = \neg v(\theta)$). Many later problems cite earlier ones, as the ground rules allow.

Verdict summary (full arguments below)

Problem 1:	(a) T (b) T (c) F (d) T (e) T (f) F (g) T
Problem 6(e):	$((p \to q) \vee (q \to p))$ taut.; $(p \leftrightarrow \neg p)$ contra.; $((p \vee q) \to p)$ neither; De Morgan taut.
Problem 7:	(a) 2^{n-2} (b) 2^{n-3} (c) 2^{n-2} (d) 3
Problem 11:	(a) F (b) F (c) T (d) F (e) T (f) F (g) T (h) T
Problem 12:	(a) T (b) T (c) F (d) T (e) T (f) F
Problem 13:	(a) I (b) III (c) I (d) II (e) II (f) III (g) existence half

Part I — Reading Theorem 2.2

Problem 1 — Reading the theorem precisely

Assume $P \neq \varnothing$ (if $P = \varnothing$ the language is empty and every statement degenerates); fix a variable $p_0 \in P$ for use in refutations.

(a) True. This is exactly the existence half of Theorem 2.2: $w = \overline{v}$ is a truth assignment with $\overline{v}(p) = v(p)$ for all $p \in P$.

(b) True. This is the uniqueness half of Theorem 2.2: at most one truth assignment agrees with v on P. (Together with (a): exactly one.)

(c) False. The condition constrains w only on the variables in P; on every compound formula w is unconstrained. Take $w_1, w_2 \colon \mathrm{Form}(P, S) \to \{\mathrm{T}, \mathrm{F}\}$ both agreeing with v on every variable, but with $w_1(\neg p_0) = \mathrm{T}$ and $w_2(\neg p_0) = \mathrm{F}$ (define them arbitrarily, but differently, here; e.g. set each equal to $\overline{v}$ except at $\neg p_0$). Both satisfy "$w(p) = v(p)$ for all $p \in P$" since $\neg p_0$ is not a variable, and $w_1 \neq w_2$. So there is more than one such function; "exactly one" fails.

(d) True. *Existence of v:* put $v = w{\restriction}P$ (the restriction of w to the variables). Then w is a truth assignment agreeing with v on P, so by the uniqueness half of Theorem 2.2, $w = \overline{v}$. *Uniqueness of v:* if $\overline{v} = w = \overline{v'}$ then for each $p \in P$, $v(p) = \overline{v}(p) = w(p) = \overline{v'}(p) = v'(p)$, so $v = v'$.

(e) True. Both u and w are truth assignments extending the common function $v := u{\restriction}P = w{\restriction}P$. By the uniqueness half of Theorem 2.2 there is only one such extension, so $u = w$; in particular $u(\varphi) = w(\varphi)$ for every φ. (This is the book's own derivation of "a truth assignment is determined by its values on the variables"; it is also Problem 8(a) taken over the full variable set.)

(f) False. Agreeing with a truth assignment on the variables does not force w to respect the truth tables at compound formulas. Let $u = \overline{v}$ where $v(p_0) = \mathrm{T}$, so $u(\neg p_0) = \mathrm{F}$. Define w to equal u

everywhere except $w(\neg p_0) = \mathrm{T}$. Then w is a genuine function $\mathrm{Form}(P, S) \to \{\mathrm{T}, \mathrm{F}\}$ agreeing with u on every variable (they agree at p_0 and at all other variables). But w violates the $\neg$-clause at $\neg p_0$: respecting $\neg$ would require $w(\neg p_0) = \neg w(p_0) = \neg \mathrm{T} = \mathrm{F}$, whereas $w(\neg p_0) = \mathrm{T}$. So w is not a truth assignment.

(g) True. For each variable p, the formula $\neg p$ has length 1, so by hypothesis $u(\neg p) = w(\neg p)$. Since u, w respect $\neg$, this says $\neg u(p) = \neg w(p)$, hence $u(p) = w(p)$. Thus u and w agree on every variable, so by (e) (equivalently the uniqueness half of Theorem 2.2) $u = w$.

Problem 2 — Extensions that are not truth assignments

Here $P = \{p\}$, $v(p) = \mathrm{T}$.

(a) Let f be the constant function T (so $f(\chi) = \mathrm{T}$ for every χ). Then $f(p) = \mathrm{T}$, but f violates the $\neg$-clause at $\neg p$: $f(\neg p) = \mathrm{T}$ while respecting $\neg$ would demand $\neg f(p) = \mathrm{F}$. Let g agree with the genuine assignment $\overline{v}$ everywhere except $g\big((p \wedge p)\big) = \mathrm{F}$. Then $g(p) = \overline{v}(p) = \mathrm{T}$, but g violates the $\wedge$-clause at $(p \wedge p)$: since $g(p) = \mathrm{T}$, respecting $\wedge$ would require $g\big((p \wedge p)\big) = \mathrm{T} \wedge \mathrm{T} = \mathrm{T}$, whereas it is F. Finally $f \neq g$: they disagree at $\neg p$ (indeed $g(\neg p) = \overline{v}(\neg p) = \mathrm{F} \neq \mathrm{T} = f(\neg p)$).

(b) $\mathrm{Form}(\{p\}, S)$ is infinite: $p, \neg p, \neg\neg p, \neg\neg\neg p, \ldots$ are pairwise distinct formulas. For each $k \geq 1$ define $f_k \colon \mathrm{Form}(\{p\}, S) \to \{\mathrm{T}, \mathrm{F}\}$ by $f_k(p) = \mathrm{T}$, $f_k(\neg^k p) = \mathrm{T}$, and $f_k(\chi) = \mathrm{F}$ for all other χ. Each f_k extends v (value T at p), and $f_j \neq f_k$ for $j \neq k$ (they differ at $\neg^j p$). So there are infinitely many functions extending v. (Almost none are truth assignments; the problem asks only for functions.)

(c) Exactly one, by the uniqueness half of Theorem 2.2. The freedom exhibited in (b) is freedom among arbitrary functions; requiring the extension to respect the truth tables collapses it to a single assignment.

(d) Let $\mathcal{A}$ be the set of truth assignments on $\mathrm{Form}(P, S)$ and $\mathcal{V}$ the set of functions $P \to \{\mathrm{T}, \mathrm{F}\}$. Define $\rho \colon \mathcal{A} \to \mathcal{V}$ by $\rho(w) = w \restriction P$ and $E \colon \mathcal{V} \to \mathcal{A}$ by $E(v) = \overline{v}$ (well-defined into $\mathcal{A}$ by the existence half of Theorem 2.2). $\rho \circ E = \mathrm{id}$: $\rho(E(v)) = \overline{v} \restriction P = v$, since $\overline{v}(p) = v(p)$ for all $p \in P$. $E \circ \rho = \mathrm{id}$: $E(\rho(w)) = \overline{w \restriction P}$; but w is a truth assignment extending $w \restriction P$, so by the uniqueness half of Theorem 2.2 the unique such extension is w itself, i.e. $\overline{w \restriction P} = w$. Being mutually inverse, ρ and E are bijections. Hence if $|P| = n$ then $|\mathcal{A}| = |\mathcal{V}| = |\{\mathrm{T}, \mathrm{F}\}|^{|P|} = 2^n$ (cf. Exercise 2.19).

Problem 3 — What unique readability is for

σ is the bracket-free string $p \wedge q \vee r$.

(a) Two construction trees:

$$T_1 : \ (p \wedge q) \vee r \qquad (\text{root } \vee, \text{ children } p \wedge q \text{ and } r),$$

$$T_2 : \ p \wedge (q \vee r) \qquad (\text{root } \wedge, \text{ children } p \text{ and } q \vee r).$$

As *strings* both spell $p \wedge q \vee r$: concatenating $p \wedge q$, $\vee$, r gives σ, and so does concatenating p, $\wedge$, $q \vee r$. Same string, two different trees.

(b) With $v(p) = \mathrm{F}$, $v(q) = v(r) = \mathrm{T}$:

$$T_1 : \quad p \wedge q \mapsto \mathrm{F} \wedge \mathrm{T} = \mathrm{F}, \qquad \mathrm{F} \vee r \mapsto \mathrm{F} \vee \mathrm{T} = \mathrm{T};$$

$$T_2 : \quad q \vee r \mapsto \mathrm{T} \vee \mathrm{T} = \mathrm{T}, \qquad p \wedge \mathrm{T} \mapsto \mathrm{F} \wedge \mathrm{T} = \mathrm{F}.$$

So the same string receives value T along T_1 and F along T_2. The recursive clauses therefore do not determine a single value for σ: there is no well-defined function from pseudo-formulas to $\{\mathrm{T}, \mathrm{F}\}$ produced by these clauses.

(c) The failing sentence is the one asserting the principal connective in the recursive step: Goldrei writes that a formula of length $n+1$ "has a principal connective *(which is unique)*, so is one of the forms $\neg\theta$, $(\theta \wedge \psi), \dots$ where θ and ψ are of length $\leq n$," and then defines the value "using the appropriate row of the truth table for its principal connective." For pseudo-formulas this uniqueness fails: σ has *two* principal connectives ($\vee$ via T_1, $\wedge$ via T_2) and two decompositions into (θ, ψ). So "the appropriate row for its principal connective" is not well specified — there are two, giving two candidate values. In genuine $\mathrm{Form}(P, S)$, unique readability guarantees exactly one principal connective and one decomposition, so the recursive rule has a single output at each step.

(d) What is threatened is the **well-definedness** of $\overline{v}$ as a function. Producing *a* value is not the issue (each tree yields one); and uniqueness in the sense of Theorem 2.2 — at most one *truth assignment* — is a separate, downstream question. Part (b) shows the *rule* "$\overline{v}(\sigma) :=$ the value from the decomposition of σ" fails to name one value, because σ has more than one decomposition with differing values. The three notions line up as a dependency chain: unique readability makes the recursive clauses well-defined (each formula has a unique immediate decomposition, so the rule has a unique output); well-definedness is what lets $\overline{v}$ be a function at all; and only once $\overline{v}$ is a function can one ask whether it exists as a truth assignment and whether it is the unique one. Removing unique readability breaks the very first link, which is what (b) exhibits.

Problem 4 — Adding a connective

(a) *v respects the truth table of $\oplus$* means: for all $\theta, \psi \in \mathrm{Form}(P, S_\oplus)$,

$$v\big((\theta \oplus \psi)\big) = \begin{cases} \mathrm{T}, & \text{if exactly one of } v(\theta), v(\psi) \text{ is T}, \\ \mathrm{F}, & \text{if } v(\theta) = v(\psi). \end{cases}$$

(b) *Analogue of Theorem 2.2.* Let P be a set of propositional variables, let $S_\oplus = \{\neg, \wedge, \vee, \rightarrow, \leftrightarrow, \oplus\}$, and let $v \colon P \to \{\mathrm{T}, \mathrm{F}\}$ be a function. Then there is a unique truth assignment $\overline{v} \colon \mathrm{Form}(P, S_\oplus) \to \{\mathrm{T}, \mathrm{F}\}$ with $\overline{v}(p) = v(p)$ for all $p \in P$ (where "truth assignment" now means a function respecting the truth tables of all six connectives in $S_\oplus$).

(c) *Existence (recursion on length).* One added case in the recursive step: if the principal connective is $\oplus$, so $\varphi = (\theta \oplus \psi)$ with θ, ψ of length $\leq n$, define $\overline{v}(\varphi)$ by the $\oplus$-row determined by $\overline{v}(\theta), \overline{v}(\psi)$. Everything else is unchanged. *Uniqueness (induction on length).* One added case: if u, w are truth assignments (now respecting all six tables) agreeing on the variables and $\varphi = (\theta \oplus \psi)$, then the induction hypothesis gives $u(\theta) = w(\theta)$ and $u(\psi) = w(\psi)$, and since both respect the $\oplus$-table, $u(\varphi) = w(\varphi)$. Otherwise unchanged.

(d) The facts needed about $\mathrm{Form}(P, S_\oplus)$, of the kind proved for $\mathrm{Form}(P, S)$ in §2.2, are: (i) every formula of length ≥ 1 has a *unique* principal connective, and correspondingly a unique immediate decomposition — either $\neg\theta$ for a unique θ, or $(\theta \circ \psi)$ for a unique connective $\circ \in \{\wedge, \vee, \rightarrow, \leftrightarrow, \oplus\}$ and unique θ, ψ (unique readability for the enlarged language); and (ii) those immediate subformulas have strictly smaller length. If uniqueness of the principal connective failed, then in the existence proof the step "define $\overline{v}(\varphi)$ by the table for its principal connective" would be ambiguous exactly as in Problem 3 — a formula might decompose both as $(\theta \oplus \psi)$ and as $(\theta' \vee \psi')$, giving two candidate values, so $\overline{v}$ would not be well-defined; and in the uniqueness proof the case split on the principal connective would be neither exhaustive nor exclusive, so the inductive step would fail to pin down $u(\varphi)$.

Part II — Computing with truth assignments

Problem 5 — Bottom-up evaluation

Here $w(p) = \mathrm{F}$, $w(q) = \mathrm{T}$, $w(r) = \mathrm{T}$.

(a) $w\big(((\neg p \vee q) \to (r \wedge \neg q))\big)$:

$$\neg p = \mathrm{T}, \quad (\neg p \vee q) = \mathrm{T} \vee \mathrm{T} = \mathrm{T}, \quad \neg q = \mathrm{F}, \quad (r \wedge \neg q) = \mathrm{T} \wedge \mathrm{F} = \mathrm{F},$$

$$\big((\neg p \vee q) \to (r \wedge \neg q)\big) = \mathrm{T} \to \mathrm{F} = \boxed{\mathrm{F}}.$$

(b) $w\big(((p \to (q \to r)) \leftrightarrow \neg(p \vee \neg r))\big)$:

$$(q \to r) = \mathrm{T} \to \mathrm{T} = \mathrm{T}, \quad (p \to (q \to r)) = \mathrm{F} \to \mathrm{T} = \mathrm{T},$$

$$\neg r = \mathrm{F}, \quad (p \vee \neg r) = \mathrm{F} \vee \mathrm{F} = \mathrm{F}, \quad \neg(p \vee \neg r) = \mathrm{T},$$

$$\big((p \to (q \to r)) \leftrightarrow \neg(p \vee \neg r)\big) = \mathrm{T} \leftrightarrow \mathrm{T} = \boxed{\mathrm{T}}.$$

(c) Determined; value T. We have $((q \wedge s) \vee r)$ with $w(r) = \mathrm{T}$, and $x \vee \mathrm{T} = \mathrm{T}$ for both values of x. So whatever $w(s)$ is, $(q \wedge s) \vee r = \mathrm{T}$. Every w consistent with the data gives T.

(d) Not determined. Here $(q \wedge s) \wedge r$; with $w(q) = w(r) = \mathrm{T}$ this equals $(\mathrm{T} \wedge w(s)) \wedge \mathrm{T} = w(s)$, which is unconstrained. Two truth assignments consistent with the data:

$$w_1 : \ (p, q, r, s) \mapsto (\mathrm{F}, \mathrm{T}, \mathrm{T}, \mathrm{T}) \ \Rightarrow \ (q \wedge s) \wedge r = (\mathrm{T} \wedge \mathrm{T}) \wedge \mathrm{T} = \mathrm{T},$$

$$w_2 : \ (p, q, r, s) \mapsto (\mathrm{F}, \mathrm{T}, \mathrm{T}, \mathrm{F}) \ \Rightarrow \ (q \wedge s) \wedge r = (\mathrm{T} \wedge \mathrm{F}) \wedge \mathrm{T} = \mathrm{F}.$$

Both exist as truth assignments by the existence half of Theorem 2.2 (they are specified by consistent values on the variables), and they give different values.

(e) Only $w(p) = \mathrm{T}$ is known.

- $(p \vee q)$: $\mathrm{T} \vee w(q) = \mathrm{T}$ always. **Determined**, value T.
- $(p \wedge q)$: $\mathrm{T} \wedge w(q) = w(q)$, unconstrained. **Not determined** ($w(q) = \mathrm{T}$ gives T; $w(q) = \mathrm{F}$ gives F).
- $(q \to p)$: $w(q) \to \mathrm{T} = \mathrm{T}$ always. **Determined**, value T.
- $(q \vee \neg q)$: $w(q) \vee \neg w(q) = \mathrm{T}$ always. **Determined**, value T (indeed independent of p as well).
- $(q \wedge \neg q)$: $w(q) \wedge \neg w(q) = \mathrm{F}$ always. **Determined**, value F.

Problem 6 — Truth tables, concrete and schematic

(a) Table of $((p \vee \neg q) \to (q \wedge r))$:

p	q	r	$p \vee \neg q$	$q \wedge r$	$((p \vee \neg q) \to (q \wedge r))$
T	T	T	T	T	T
T	T	F	T	F	F
T	F	T	T	F	F
T	F	F	T	F	F
F	T	T	F	T	T
F	T	F	F	F	T
F	F	T	T	F	F
F	F	F	T	F	F

(True in exactly the three rows TTT, FTT, FTF; note the two T's in rows $5, 6$ come from a *false* antecedent.)

(b) Table of $((\varphi \to \psi) \to \varphi)$ in the values of φ, ψ:

φ	ψ	$(\varphi \to \psi)$	$((\varphi \to \psi) \to \varphi)$
T	T	T	T
T	F	F	T
F	T	T	F
F	F	T	F

The last column equals the φ-column in every row: $((\varphi \to \psi) \to \varphi)$ takes the same value as φ under every truth assignment.

(c) Realizing $(\varphi, \psi) = (\mathrm{F}, \mathrm{T})$: take $\varphi_0 = p$, $\psi_0 = q$, and w with $w(p) = \mathrm{F}$, $w(q) = \mathrm{T}$. Then $w(\varphi_0) = \mathrm{F}$, $w(\psi_0) = \mathrm{T}$, and

$$w\big(((p \to q) \to p)\big): \quad (p \to q) = \mathrm{F} \to \mathrm{T} = \mathrm{T}, \quad (\mathrm{T} \to p) = \mathrm{T} \to \mathrm{F} = \mathrm{F}.$$

So the value is F, matching the row (F, T) of (b) (whose entry equals $\varphi = \mathrm{F}$).

(d) Put $\psi := \neg\varphi$. Under any w, $w(\psi) = \neg w(\varphi)$, so the pair $(w(\varphi), w(\psi))$ is always (T, F) or (F, T): the rows (T, T) and (F, F) become **unrealizable**, since they would require $w(\varphi) = w(\psi) = w(\varphi) = \neg w(\varphi)$, impossible. On the surviving rows, and indeed on all of them, (b) gives $((\varphi \to \psi) \to \varphi) = \varphi$; hence $((\varphi \to \neg\varphi) \to \varphi)$ takes the same value as φ under every w. Therefore it is a tautology iff φ is true under every w, i.e. iff φ is a tautology; and a contradiction iff φ is false under every w, i.e. iff φ is a contradiction.

(e) *Classifications.*

- $((p \to q) \vee (q \to p))$ — **tautology.** If both disjuncts were false under some w, then $(p \to q) = \mathrm{F}$ gives $w(p) = \mathrm{T}, w(q) = \mathrm{F}$, while $(q \to p) = \mathrm{F}$ gives $w(q) = \mathrm{T}, w(p) = \mathrm{F}$; incompatible. So at least one disjunct is T under every w.
- $(p \leftrightarrow \neg p)$ — **contradiction.** $w(p) \leftrightarrow \neg w(p)$ is $\mathrm{T} \leftrightarrow \mathrm{F} = \mathrm{F}$ or $\mathrm{F} \leftrightarrow \mathrm{T} = \mathrm{F}$.
- $((p \vee q) \to p)$ — **neither.** $w(p) = \mathrm{T}$ gives $\mathrm{T} \to \mathrm{T} = \mathrm{T}$; $w(p) = \mathrm{F}, w(q) = \mathrm{T}$ gives $\mathrm{T} \to \mathrm{F} = \mathrm{F}$.
- $(\neg(p \wedge q) \leftrightarrow (\neg p \vee \neg q))$ — **tautology** (De Morgan). Checking the four rows: $(\mathrm{T}, \mathrm{T}): \mathrm{F} \leftrightarrow \mathrm{F} = \mathrm{T}$; $(\mathrm{T}, \mathrm{F}): \mathrm{T} \leftrightarrow \mathrm{T} = \mathrm{T}$; $(\mathrm{F}, \mathrm{T}): \mathrm{T} \leftrightarrow \mathrm{T} = \mathrm{T}$; $(\mathrm{F}, \mathrm{F}): \mathrm{T} \leftrightarrow \mathrm{T} = \mathrm{T}$.

Problem 7 — Counting assignments

Values on the "free" variables (those not named in the formula) may be anything, contributing a factor $2^{(\text{number free})}$; the named variables are pinned.

(a) $(p_1 \wedge \neg p_2)$ is true iff $p_1 = \mathrm{T}$ and $p_2 = \mathrm{F}$; $p_3, \ldots, p_n$ free. Count $= 2^{n-2}$.

(b) $\neg((p_1 \vee p_2) \vee p_3)$ is true iff $(p_1 \vee p_2 \vee p_3) = \mathrm{F}$ iff $p_1 = p_2 = p_3 = \mathrm{F}$; the rest free. Count $= 2^{n-3}$.

(c) $\neg(p_1 \to p_2)$ is true iff $(p_1 \to p_2) = \mathrm{F}$ iff $p_1 = \mathrm{T}, p_2 = \mathrm{F}$; the rest free. Count $= 2^{n-2}$.

(d) $((p_1 \vee p_2) \wedge p_3)$ with $n = 3$: need $p_3 = \mathrm{T}$ and $(p_1 \vee p_2) = \mathrm{T}$. Fixing $p_3 = \mathrm{T}$, the condition $p_1 \vee p_2 = \mathrm{T}$ excludes only $p_1 = p_2 = \mathrm{F}$, leaving 3 of the 4 patterns of (p_1, p_2). Count $= 3$.

Part III — Inductions on the length of a formula

Problem 8 — The relevance lemma

(a) Claim. If u, w are truth assignments and $u(q) = w(q)$ for every $q \in \mathrm{vars}(\varphi)$, then $u(\varphi) = w(\varphi)$.

Proof by induction on the length of φ. Induction hypothesis: for every formula θ of length $\leq n$, if u, w agree on every variable in $\mathrm{vars}(\theta)$, then $u(\theta) = w(\theta)$.

Base (φ of length 0). Then φ is a variable p, and $\mathrm{vars}(\varphi) = \{p\}$; the hypothesis gives $u(p) = w(p)$, i.e. $u(\varphi) = w(\varphi)$.

Inductive step (φ of length $n+1$). By unique readability φ has one principal connective.

- $\varphi = \neg\theta$. Then $\mathrm{vars}(\varphi) = \mathrm{vars}(\theta)$, so u, w agree on $\mathrm{vars}(\theta)$; as θ has length n, the IH gives $u(\theta) = w(\theta)$. Since u, w *respect* $\neg$ (used here), $u(\neg\theta) = \neg u(\theta) = \neg w(\theta) = w(\neg\theta)$.
- $\varphi = (\theta \wedge \psi)$. Then $\mathrm{vars}(\varphi) = \mathrm{vars}(\theta) \cup \mathrm{vars}(\psi)$, so u, w agree on each of $\mathrm{vars}(\theta), \mathrm{vars}(\psi)$; both have length $\leq n$, so the IH gives $u(\theta) = w(\theta)$ and $u(\psi) = w(\psi)$. Since u, w *respect* $\wedge$ (used here), $u((\theta \wedge \psi)) = u(\theta) \wedge u(\psi) = w(\theta) \wedge w(\psi) = w((\theta \wedge \psi))$.
- $\varphi = (\theta \to \psi)$. As above u, w agree on $\mathrm{vars}(\theta), \mathrm{vars}(\psi)$; the IH gives $u(\theta) = w(\theta)$, $u(\psi) = w(\psi)$; since u, w *respect* $\to$ (used here), the value $u((\theta \to \psi))$, read off the $\to$-table from $(u(\theta), u(\psi)) = (w(\theta), w(\psi))$, equals $w((\theta \to \psi))$.
- The cases $\vee, \leftrightarrow$ are identical, using the respective tables.

This closes the induction. □

(b) *Exercise 2.18.* Suppose v, v' are truth assignments agreeing at every variable except possibly p (that is, $v(q) = v'(q)$ for all $q \neq p$), and let φ be a formula (over $\{\neg, \wedge, \vee\}$, as in the book) in which p does not occur. Every $q \in \mathrm{vars}(\varphi)$ satisfies $q \neq p$, so v, v' agree on all of $\mathrm{vars}(\varphi)$; by (a), $v(\varphi) = v'(\varphi)$. The analogue for the full set S is immediate, since (a) was proved for S. (Indeed (a) is stronger: the two assignments may differ at *any* set of variables disjoint from $\mathrm{vars}(\varphi)$, not just at a single p.)

(c) *Soundness of truth tables.* Write $V = \mathrm{vars}(\varphi) \subseteq \{p_1, \ldots, p_n\}$.

The bridge. Part (a) as stated compares two assignments on a common domain. For soundness we need the mild variant: *if u is a truth assignment on* $\mathrm{Form}(P, S)$ *and u' a truth assignment on* $\mathrm{Form}(P', S)$ *with* $\mathrm{vars}(\varphi) \subseteq P \cap P'$, *and $u(q) = u'(q)$ for every $q \in \mathrm{vars}(\varphi)$, then $u(\varphi) = u'(\varphi)$.* The proof is *verbatim* the induction in (a): at each node the current subformula uses only variables in $\mathrm{vars}(\varphi) \subseteq P \cap P'$, hence lies in both $\mathrm{Form}(P, S)$ and $\mathrm{Form}(P', S)$ and the IH applies; the argument never uses that the two domains coincide, only that every subformula of φ lives in both. This is the point the problem flags: "agree on the variables of φ" is the right hypothesis even across different domains.

Soundness. ($\Rightarrow$) If φ is a tautology — true under every truth assignment defined on any variable set containing $\mathrm{vars}(\varphi)$ — then in particular it is true under each of the 2^n assignments on $\mathrm{Form}(\{p_1, \ldots, p_n\}, S)$.

($\Leftarrow$) Suppose all 2^n assignments on $\mathrm{Form}(\{p_1, \ldots, p_n\}, S)$ make φ true. Let w be an arbitrary truth assignment on some $\mathrm{Form}(P, S)$ with $V \subseteq P$. Define $g\colon \{p_1, \ldots, p_n\} \to \{\mathrm{T}, \mathrm{F}\}$ by $g(p_i) = w(p_i)$ when $p_i \in P$ and $g(p_i) = \mathrm{T}$ otherwise (the arbitrary choice affects only $p_i \notin V$, since $p_i \in V \Rightarrow p_i \in P$). By the existence half of Theorem 2.2, $\overline{g}$ is one of the 2^n assignments on $\mathrm{Form}(\{p_1, \ldots, p_n\}, S)$, so $\overline{g}(\varphi) = \mathrm{T}$ by hypothesis. Now $\overline{g}$ and w agree on every $q \in V = \mathrm{vars}(\varphi)$ (for such $q = p_i \in P$ we have $\overline{g}(p_i) = g(p_i) = w(p_i)$), so the cross-domain bridge gives $w(\varphi) = \overline{g}(\varphi) = \mathrm{T}$. As w was arbitrary, φ is a tautology.

(If some $p_i \notin V$, the 2^n assignments simply test each genuinely relevant pattern $2^{n-|V|}$ times; the criterion is unaffected.)

Problem 9 — Negating the variables: Exercise 2.28 extended

Recall v^* is the assignment with $v^*(p) = \neg v(p)$ on variables (a truth assignment by Theorem 2.2), and φ^* is defined by the stated recursion (each variable negated, connectives unchanged).

(a) Claim. $v(\varphi) = v^*(\varphi^*)$ for every truth assignment v and formula φ. Fix v (hence v^*) and induct on the length of φ; IH: this holds for all formulas of length $\leq n$.

Base ($\varphi = p$). $\varphi^* = \neg p$, so

$$v^*(\varphi^*) = v^*(\neg p) = \neg v^*(p) = \neg(\neg v(p)) = v(p) = v(\varphi),$$

using that v^* respects $\neg$ and the definition $v^*(p) = \neg v(p)$.

Step.

- $\varphi = \neg\theta$: $\varphi^* = \neg\theta^*$, so $v^*(\varphi^*) = \neg v^*(\theta^*) = \neg v(\theta) = v(\neg\theta) = v(\varphi)$, by the $\neg$-clause (for v^* and for v) and the IH on θ.
- $\varphi = (\theta \to \psi)$: $\varphi^* = (\theta^* \to \psi^*)$. Both $v^*(\varphi^*)$ and $v(\varphi)$ are read off the *same* $\to$-table, the former from $(v^*(\theta^*), v^*(\psi^*))$ and the latter from $(v(\theta), v(\psi))$; by the IH these input pairs coincide, so the outputs coincide.
- $\varphi = (\theta \leftrightarrow \psi)$: identical, using the $\leftrightarrow$-table.
- $\wedge, \vee$ similar.

(The key structural point: $*$ negates variables but leaves each binary connective *unchanged*, so at each step the two sides use the identical truth table.) $\square$

(b) $(v^*)^*$ is the assignment with $(v^*)^*(p) = \neg v^*(p) = \neg(\neg v(p)) = v(p)$ for every variable p. So $(v^*)^*$ and v are truth assignments agreeing on all variables; by the uniqueness half of Theorem 2.2 they are equal as functions on $\mathrm{Form}(P, S)$, i.e. $(v^*)^* = v$. Hence $v \mapsto v^*$ is its own inverse, so a bijection of the set of truth assignments onto itself.

(c) φ tautology $\iff$ φ^* tautology. ($\Rightarrow$) Let w be an arbitrary truth assignment; we show $w(\varphi^*) = \mathrm{T}$. Put $v := w^*$ (a truth assignment). By (a), $v(\varphi) = v^*(\varphi^*)$, and by (b) $v^* = (w^*)^* = w$; so $w(\varphi^*) = v(\varphi)$. Since φ is a tautology, $v(\varphi) = \mathrm{T}$, hence $w(\varphi^*) = \mathrm{T}$. As w was arbitrary, φ^* is a tautology. ($\Leftarrow$) Let w be arbitrary; by (a) with $v := w$, $w(\varphi) = w^*(\varphi^*)$; since φ^* is a tautology, $w^*(\varphi^*) = \mathrm{T}$, so $w(\varphi) = \mathrm{T}$.

The quantifier point the problem asks for: in ($\Rightarrow$) we need $w(\varphi^*)$ for *arbitrary* w, and (a) only directly gives values of the form $v^*(\varphi^*)$. This suffices precisely because $*$ is *onto* (indeed a bijection, by (b)): every assignment w equals v^* for some v — namely $v = w^*$ — so the universally quantified identity in (a) reaches every w.

(d) φ contradiction $\iff$ φ^* contradiction, no new induction. By Exercise 2.26, φ is a contradiction iff $\neg\varphi$ is a tautology. Also $(\neg\varphi)^* = \neg(\varphi^*)$ by the definition of $*$. Therefore

$$\begin{aligned} \varphi \text{ contradiction} &\iff \neg\varphi \text{ tautology} \overset{(c)}{\iff} (\neg\varphi)^* \text{ tautology} \\ &\iff \neg(\varphi^*) \text{ tautology} \iff \varphi^* \text{ contradiction}, \end{aligned}$$

the middle step applying (c) to $\neg\varphi$ and the last again by Exercise 2.26.

(e) True: $v(\varphi^*) = v^*(\varphi)$ for all v, φ, with no new induction. Apply (a) to the assignment $u := v^*$: $u(\varphi) = u^*(\varphi^*)$, i.e. $v^*(\varphi) = (v^*)^*(\varphi^*) = v(\varphi^*)$, using $(v^*)^* = v$ from (b). Reading the identity backwards, $v(\varphi^*) = v^*(\varphi)$.

Problem 10 — $\star$ The substitution lemma

(a) By unique readability every formula falls under exactly one clause of the recursion: it is the variable p, or a variable $q \neq p$, or $\neg\theta$ for a unique θ, or $(\theta \circ \psi)$ for a unique connective and unique θ, ψ. Since the clauses never compete, $\varphi[\chi/p]$ is specified unambiguously — the same definition-by-recursion justified by the existence half of Theorem 2.2.

(b) Claim. $v(\varphi[\chi/p]) = u(\varphi)$, where u is the truth assignment with $u(p) = v(\chi)$ and $u(q) = v(q)$ for $q \neq p$. (Such u exists and is unique by Theorem 2.2, applied to the function on variables sending $p \mapsto v(\chi)$ and $q \mapsto v(q)$.) Fix v (hence u) and induct on the length of φ; IH: the identity holds for formulas of length $\leq n$.

Base, two subcases.

- $\varphi = p$: $\varphi[\chi/p] = \chi$, so $v(\varphi[\chi/p]) = v(\chi) = u(p) = u(\varphi)$.
- $\varphi = q$, $q \neq p$: $\varphi[\chi/p] = q$, so $v(\varphi[\chi/p]) = v(q) = u(q) = u(\varphi)$.

Step.

- $\varphi = \neg\theta$: $\varphi[\chi/p] = \neg(\theta[\chi/p])$, so $v(\varphi[\chi/p]) = \neg v(\theta[\chi/p]) = \neg u(\theta) = u(\neg\theta) = u(\varphi)$, by the $\neg$-clause and the IH.
- $\varphi = (\theta \to \psi)$: then $\varphi[\chi/p] = (\theta[\chi/p] \to \psi[\chi/p])$, and by the IH applied to θ and ψ,

$$\big(v(\theta[\chi/p]),\, v(\psi[\chi/p])\big) = \big(u(\theta),\, u(\psi)\big).$$

  Reading both $v(\varphi[\chi/p])$ and $u((\theta \to \psi))$ off the $\to$-table from this common pair gives $v(\varphi[\chi/p]) = u((\theta \to \psi)) = u(\varphi)$.
- $\wedge, \vee, \leftrightarrow$ similar.

□

(c) If φ is a tautology then, for every truth assignment v, $v(\varphi[\chi/p]) = u(\varphi) = \mathrm{T}$ by (b) (with u the assignment it names, itself a genuine truth assignment). As v was arbitrary, $\varphi[\chi/p]$ is a tautology.

(d) The converse fails. Take $\varphi = (p \vee q)$, substitute for p the formula $\chi = \neg q$. Then

$$\varphi[\chi/p] = (\neg q \vee q),$$

which is a tautology: $w(\neg q \vee q) = \neg w(q) \vee w(q) = \mathrm{T}$ for every w. But $\varphi = (p \vee q)$ is *not* a tautology: the assignment w with $w(p) = \mathrm{F}$, $w(q) = \mathrm{F}$ gives $w((p \vee q)) = \mathrm{F} \vee \mathrm{F} = \mathrm{F}$. So $\varphi[\chi/p]$ is a tautology while φ is not.

(e) By (b), the value of a formula built from φ, ψ under w equals the value of the corresponding two-variable formula under the assignment sending $p \mapsto w(\varphi)$, $q \mapsto w(\psi)$; when φ, ψ are distinct variables, Theorem 2.2 lets $(w(\varphi), w(\psi))$ be set to any of the four pairs independently, so every row of the schematic table is realized. When φ, ψ are correlated — e.g. $\psi = \neg\varphi$ as in Problem 6(d) — the pair is constrained (there $w(\psi) = \neg w(\varphi)$), so rows violating the constraint have no realizing assignment and drop out. In short, distinct metavariables need not be independently settable; only genuinely independent variables reach all rows.

Part IV — Tautologies, contradictions, counterexamples

Problem 11 — Calibration: true or false?

(a) False. Agreement at φ does not force agreement at its variables. Take $\varphi = (p \vee q)$; let $u(p) = \mathrm{T}, u(q) = \mathrm{T}$ and $w(p) = \mathrm{T}, w(q) = \mathrm{F}$. Then $u(\varphi) = \mathrm{T}$ and $w(\varphi) = \mathrm{T} \vee \mathrm{F} = \mathrm{T}$, so $u(\varphi) = w(\varphi)$, yet $q \in \mathrm{vars}(\varphi)$ and $u(q) = \mathrm{T} \neq \mathrm{F} = w(q)$. (This is the *converse* of relevance, and it is false.)

(b) False. Any truth assignment respects $\wedge$: $v((p \wedge q)) = v(p) \wedge v(q)$. If $v(q) = \mathrm{F}$ then $v(p) \wedge \mathrm{F} = \mathrm{F} \neq \mathrm{T}$. So no truth assignment does this.

(c) True. A mere function need not respect $\wedge$. Define f with $f((p \wedge q)) = \mathrm{T}$, $f(q) = \mathrm{F}$, and f arbitrary (say T) elsewhere. This is a well-defined function with the two required values. (Contrast (b): the $\wedge$-clause is exactly what a function may flout but an assignment may not.)

(d) False. Take $\varphi = p$. Then p is not a tautology ($w(p) = \text{F}$ falsifies it), and $\neg p$ is not a tautology either ($w(p) = \text{T}$ gives $\neg p = \text{F}$). So "φ not a tautology" does not make $\neg\varphi$ one: p is merely *not forced true*, which is weaker than $\neg p$ being *forced true*.

(e) True. By Exercise 2.26, $\neg\varphi$ is a contradiction iff $\neg\neg\varphi$ is a tautology iff φ is a tautology. Contrapositively, if φ is not a tautology then $\neg\varphi$ is not a contradiction.

(f) False. Take $\varphi = p$, $\psi = \neg p$: neither is a tautology (p fails at $w(p) = \text{F}$; $\neg p$ fails at $w(p) = \text{T}$), yet $(\varphi \vee \psi) = (p \vee \neg p)$ is a tautology.

(g) True. For every w, $w((\varphi \wedge \psi)) = w(\varphi) \wedge w(\psi)$; the $\wedge$-table yields T only when both conjuncts are T. So if $(\varphi \wedge \psi)$ is a tautology, $w(\varphi) = \text{T}$ for every w, i.e. φ is a tautology (and so is ψ).

(h) True. "φ is a contradiction" means $w(\varphi) = \text{F}$ for every w; its negation is that some w has $w(\varphi) \neq \text{F}$, hence (values being only T, F) $w(\varphi) = \text{T}$. So a non-contradiction is made true by some assignment.

Problem 12 — Around modus ponens: Exercise 2.27 revisited

(a) True. For every w, $w(\varphi) = \text{T}$ and $w(\psi) = \text{T}$ (both tautologies), so $w((\varphi \wedge \psi)) = \text{T} \wedge \text{T} = \text{T}$.

(b) True (this is Problem 11(g), both conjuncts). For every w, $w(\varphi) \wedge w(\psi) = \text{T}$ forces $w(\varphi) = w(\psi) = \text{T}$; so both are tautologies.

(c) False. Take $\varphi = p$, $\psi = \neg p$. Then $(\varphi \vee \psi) = (p \vee \neg p)$ is a tautology, but $\varphi = p$ is not ($w(p) = \text{F}$) and $\psi = \neg p$ is not ($w(p) = \text{T}$). The disjunction is *forced* true, yet neither disjunct is forced true — the 2.27(c)-style trap.

(d) True. For every w: ψ a contradiction gives $w(\psi) = \text{F}$, and $(\varphi \to \psi)$ a tautology gives $w((\varphi \to \psi)) = \text{T}$. In the $\to$-table a value T with consequent F forces the antecedent F; so $w(\varphi) = \text{F}$. Thus φ is a contradiction.

(e) True. For every w: $w(\varphi) = \text{T}$ (φ a tautology) and $w(\varphi) = w(\psi)$ (from $(\varphi \leftrightarrow \psi)$ a tautology), so $w(\psi) = \text{T}$. Hence ψ is a tautology.

(f) False. Take $\varphi = \psi = p$. Then $(\varphi \leftrightarrow \psi) = (p \leftrightarrow p)$ is a tautology ($w(p) \leftrightarrow w(p) = \text{T}$ always), but $\varphi = p$ is not a tautology ($w(p) = \text{F}$). The biconditional is forced true because the two sides are *identical*, not because either is a tautology.

Problem 13 — Forced, impossible, or independent

(a) (I): H forces C. Suppose ψ is a tautology. For every w, $w(\psi) = \text{T}$, so $w((\varphi \to \psi)) = w(\varphi) \to \text{T} = \text{T}$. Hence $(\varphi \to \psi)$ is a tautology — for *every* φ. Every instance of H satisfies C.

(b) (III): C is independent of H. (This is Exercise 2.27(c).) We give two instances of H — both $(\varphi \to \psi)$ and ψ tautologies.

- *C holds.* $\varphi = (p \vee \neg p)$, $\psi = (q \vee \neg q)$. Check H: ψ is a tautology; and $(\varphi \to \psi)$ is a tautology since its consequent ψ is (for every w, $w((\varphi \to \psi)) = w(\varphi) \to \text{T} = \text{T}$). Check C: $\varphi = (p \vee \neg p)$ is a tautology. So this instance satisfies C.
- *C fails.* $\varphi = p$, $\psi = (q \vee \neg q)$. Check H: ψ is a tautology; and $(\varphi \to \psi) = (p \to (q \vee \neg q))$ is a tautology (consequent always T). Check C: $\varphi = p$ is *not* a tautology, since $w(p) = \text{F}$ gives $w(p) = \text{F}$. So this instance fails C.

Both satisfy H; one satisfies C and one does not, so we are in case (III). The lesson: $(\varphi \to \psi)$ a tautology says only $w(\varphi) \to w(\psi) = \text{T}$ for every w; when $w(\psi) = \text{T}$ (which holds, as ψ is a tautology) this is satisfied *whatever* $w(\varphi)$ is, so φ is left completely free. Object-level modus ponens fails to transfer to "all assignments" here.

(c) (I): H forces C. Suppose φ is a contradiction. For every w, $w(\varphi) = \text{F}$, so $w((\varphi \to \psi)) = \text{F} \to w(\psi) = \text{T}$. Hence $(\varphi \to \psi)$ is a tautology, for every ψ. (*Ex falso*: a false antecedent makes the implication vacuously true.)

(d) (II): H forces the failure of C. Suppose φ is a contradiction. For every w, $w(\varphi) = \text{F}$, so $w((\varphi \wedge \psi)) = \text{F} \wedge w(\psi) = \text{F}$. Thus $(\varphi \wedge \psi)$ is false under every assignment; since at least one assignment exists (part (g)) and no formula is both a tautology and a contradiction (part (g)), $(\varphi \wedge \psi)$ is *not* a tautology. So no instance of H satisfies C. (Same H as (c), opposite outcome: $\neg$ makes $\to$ vacuously true but $\wedge$ vacuously false.)

(e) (II): H forces the failure of C. If $(\varphi \to \psi)$ is a contradiction then for every w, $w((\varphi \to \psi)) = \text{F}$; the only $\to$-row with value F is antecedent T, consequent F, so $w(\varphi) = \text{T}$ and $w(\psi) = \text{F}$ for every w. In particular ψ is a contradiction, hence (by (g), and since an assignment exists) not a tautology. So no instance of H satisfies C. (Indeed H here is very restrictive: it forces φ to be a tautology and ψ a contradiction.)

(f) (III): C is independent of H. (This upgrades Problem 11(d).) Two instances of H — φ not a tautology.

- *C holds.* $\varphi = (p \wedge \neg p)$, a contradiction, hence not a tautology (it is false under, e.g., $w(p) = \text{T}$). Then $\neg\varphi = \neg(p \wedge \neg p)$ is a tautology by Exercise 2.26. So C holds.
- *C fails.* $\varphi = p$: not a tautology ($w(p) = \text{F}$). And $\neg\varphi = \neg p$ is not a tautology ($w(p) = \text{T}$ gives $\neg p = \text{F}$). So C fails.

Both satisfy H; C goes both ways, so (III). ("φ not forced true" is compatible both with φ being contingent — then $\neg\varphi$ is also contingent — and with φ being a contradiction — then $\neg\varphi$ is a tautology. This is exactly why (III), not (II).)

(g) No formula is both a tautology and a contradiction. Suppose φ were both. As a tautology, $w(\varphi) = \text{T}$ for every truth assignment w; as a contradiction, $w(\varphi) = \text{F}$ for every w. It remains to know that *there is* at least one truth assignment: take any function $v\colon P \to \{\text{T}, \text{F}\}$ (for instance the constant T), and by the **existence half of Theorem 2.2** it extends to a truth assignment $\overline{v}$. For that $\overline{v}$ we get $\text{T} = \overline{v}(\varphi) = \text{F}$, a contradiction. Hence no φ is both. The proof uses the *existence* half: were the space of assignments empty, "true under all" and "false under all" could hold vacuously at once, and the two notions would not be incompatible.

Problem 14 — Characterizations: Exercise 2.31 revisited

(a) Given φ a tautology: $(\varphi \to \psi)$ is a tautology iff ψ is a tautology. ($\Leftarrow$) If ψ is a tautology then $w((\varphi \to \psi)) = w(\varphi) \to \text{T} = \text{T}$ for every w (this direction needs nothing about φ). ($\Rightarrow$) Suppose $(\varphi \to \psi)$ and φ are tautologies. For every w, $w(\varphi) = \text{T}$ and $w((\varphi \to \psi)) = \text{T}$; the $\to$-row with antecedent T and value T has consequent T, so $w(\psi) = \text{T}$. Thus ψ is a tautology.

(b) Given ψ a tautology: $(\varphi \to \psi)$ is *never* a contradiction. If ψ is a tautology then $w((\varphi \to \psi)) = w(\varphi) \to \text{T} = \text{T}$ for every w, so $(\varphi \to \psi)$ is itself a tautology; being a tautology (and, by 13(g), not simultaneously a contradiction) it is not a contradiction, for any φ. Circumstances: none.

(c) Refuted: φ, ψ need not be contradictions. Take $\varphi = p$, $\psi = \neg p$. Then $(\varphi \wedge \psi) = (p \wedge \neg p)$ is a contradiction ($w(p) \wedge \neg w(p) = \text{F}$ always), but $\varphi = p$ is not ($w(p) = \text{T}$ gives $p = \text{T}$) and $\psi = \neg p$ is not ($w(p) = \text{F}$ gives $\neg p = \text{T}$). (Dual to 12(c): the conjunction forced false does not force either conjunct false.)

(d) $(\varphi \vee \psi)$ is a contradiction iff both φ and ψ are contradictions. ($\Rightarrow$) If $(\varphi \vee \psi)$ is a contradiction then $w((\varphi \vee \psi)) = \text{F}$ for every w; the only $\vee$-row with value F has both disjuncts F, so $w(\varphi) = \mathrm{F}$ and $w(\psi) = \mathrm{F}$ for every w — both are contradictions. ($\Leftarrow$) If both are contradictions then $w(\varphi) = w(\psi) = \mathrm{F}$, so $w((\varphi \vee \psi)) = \mathrm{F} \vee \mathrm{F} = \mathrm{F}$, for every w. (Contrast (c): for $\vee$ the condition really does descend to each disjunct.)

(e) **Refuted**, then characterized. Take $\varphi = p$, $\psi = \neg p$: $(\varphi \leftrightarrow \psi) = (p \leftrightarrow \neg p)$ is a contradiction ($w(p) \leftrightarrow \neg w(p) = \mathrm{F}$ always), but neither p nor $\neg p$ is a contradiction. *Characterization:* $(\varphi \leftrightarrow \psi)$ is a contradiction iff $w(\varphi) \neq w(\psi)$ for every truth assignment w (i.e. φ and ψ take opposite values everywhere; equivalently $\psi \equiv \neg\varphi$). Indeed the $\leftrightarrow$-table gives $w((\varphi \leftrightarrow \psi)) = \mathrm{F}$ exactly when $w(\varphi) \neq w(\psi)$; so $(\varphi \leftrightarrow \psi)$ is false under *every* w iff $w(\varphi) \neq w(\psi)$ for every w.

Problem 15 — Majority rule: Exercise 2.29 revisited

For a formula φ write $T(\varphi) = \{v \in S : v(\varphi) = \mathrm{T}\}$; then $|T(\varphi)| + |T(\neg\varphi)| = |S|$, and $\varphi \in D$ iff $|T(\varphi)| > |S|/2$.

(a) For each $v \in S$, $v(\neg\varphi) = \neg v(\varphi)$, so $v \in T(\varphi)$ iff $v \notin T(\neg\varphi)$; hence $|T(\neg\varphi)| = |S| - |T(\varphi)|$. Because $|S|$ is odd, $|T(\varphi)| \neq |S|/2$ (no tie), so exactly one of $|T(\varphi)| > |S|/2$ and $|T(\neg\varphi)| = |S| - |T(\varphi)| > |S|/2$ holds. Thus exactly one of $\varphi, \neg\varphi$ is in D.

(b) If $(\varphi \to \theta)$ is a tautology then every $v \in S$ has $v((\varphi \to \theta)) = \mathrm{T}$, i.e. $v(\varphi) = \mathrm{T} \Rightarrow v(\theta) = \mathrm{T}$; hence $T(\varphi) \subseteq T(\theta)$. So $|T(\theta)| \geq |T(\varphi)| > |S|/2$, giving $\theta \in D$. (Tautological implication can only enlarge the true-set.)

(c) Take the three assignments on $\{p, q\}$

$$v_1 : (p, q) = (\mathrm{T}, \mathrm{T}), \qquad v_2 : (p, q) = (\mathrm{T}, \mathrm{F}), \qquad v_3 : (p, q) = (\mathrm{F}, \mathrm{F}).$$

Majority means at least 2 of 3.

- $p \in D$: p is true under v_1, v_2 — 2 of 3. ✓
- $(p \to q) \in D$: $(p \to q)$ is true under v_1 ($\mathrm{T} \to \mathrm{T}$) and v_3 ($\mathrm{F} \to \mathrm{F} = \mathrm{T}$), false under v_2 ($\mathrm{T} \to \mathrm{F}$) — 2 of 3. ✓
- $q \notin D$: q is true only under v_1 — 1 of 3, not a majority. ✓

So $p, (p \to q) \in D$ but $q \notin D$: majority rule is not closed under modus ponens.

(d) **Refuted.** Take $\varphi = p$, $\psi = q$ and the three assignments

$$v_1 : (\mathrm{T}, \mathrm{F}), \qquad v_2 : (\mathrm{F}, \mathrm{T}), \qquad v_3 : (\mathrm{T}, \mathrm{T}).$$

Then p is true under v_1, v_3 (2 of 3), so $p \in D$; q is true under v_2, v_3 (2 of 3), so $q \in D$; but $(p \wedge q)$ is true only under v_3 (1 of 3), so $(p \wedge q) \notin D$. Hence D need not be closed under $\wedge$.

(e) Drop oddness. Take $|S| = 2$ on $\{p\}$: $v_1(p) = \mathrm{T}$, $v_2(p) = \mathrm{F}$. For $\varphi = p$, $|T(p)| = 1$, which is not $> |S|/2 = 1$, so $p \notin D$; likewise $|T(\neg p)| = 1$, so $\neg p \notin D$. Neither p nor $\neg p$ is in D, so the conclusion of (a) fails (an even split is now possible).

(f) **$|S| = 1$: D is closed under modus ponens.** Write $S = \{v\}$. A "majority" of a one-element set is that one element, so $D = \{\varphi : v(\varphi) = \mathrm{T}\}$. If $\varphi \in D$ and $(\varphi \to \theta) \in D$ then $v(\varphi) = \mathrm{T}$ and $v((\varphi \to \theta)) = \mathrm{T}$, i.e. $v(\varphi) \to v(\theta) = \mathrm{T}$; with $v(\varphi) = \mathrm{T}$ the $\to$-table forces $v(\theta) = \mathrm{T}$, so $\theta \in D$.

The step that breaks at $|S| = 3$. For a single assignment, "$\varphi \in D$" and "$(\varphi \to \theta) \in D$" are witnessed by the *same* v, which then also witnesses θ: one assignment does all the work. For $|S| = 3$, membership $\varphi \in D$ is carried by a majority M_φ (size ≥ 2) and $(\varphi \to \theta) \in D$ by a possibly *different* majority $M_{\varphi \to \theta}$ (size ≥ 2). Modus ponens applies only to assignments in $M_\varphi \cap M_{\varphi \to \theta}$ — those that make both φ and $\varphi \to \theta$ true, and hence θ true — but two 2-element majorities of a 3-element set can meet in just *one* assignment, which is not a majority. That is exactly what (c) exploits: $M_p = \{v_1, v_2\}$ and $M_{p \to q} = \{v_1, v_3\}$ meet only in $\{v_1\}$, so only v_1 is forced to make q true — 1 of 3, not enough for $q \in D$.

Part V — Capstone

Problem 16 — $\star$ Repetition-free formulas: Exercise 2.30 generalized

(a) Gluing Lemma. Let θ, ψ share no variable, and let u, w be truth assignments (say on a common $\mathrm{Form}(P, S)$ with $\mathrm{vars}(\theta) \cup \mathrm{vars}(\psi) \subseteq P$). Define x on the variables by

$$x(r) = \begin{cases} u(r), & r \in \mathrm{vars}(\theta), \\ w(r), & r \in \mathrm{vars}(\psi), \\ \mathrm{T}, & \text{otherwise.} \end{cases}$$

This is well-defined because $\mathrm{vars}(\theta) \cap \mathrm{vars}(\psi) = \varnothing$, so the first two clauses never conflict. By the *existence half of Theorem 2.2*, x extends to a truth assignment $\overline{x}$ on $\mathrm{Form}(P, S)$. Now $\overline{x}$ agrees with u on every variable of θ, so by *Problem 8(a)* (relevance), $\overline{x}(\theta) = u(\theta)$; likewise $\overline{x}$ agrees with w on every variable of ψ, so $\overline{x}(\psi) = w(\psi)$. Renaming $\overline{x}$ as x, this is the required assignment. $\square$

(b) Claim. Every repetition-free formula of $\mathrm{Form}(P, S)$ is made true by some assignment and false by some assignment. Induct on the length; IH: this holds for all repetition-free formulas of length $\leq n$.

Base ($\chi = p$). A variable is repetition-free; $p \mapsto \mathrm{T}$ makes it true and $p \mapsto \mathrm{F}$ makes it false (both extend to assignments by Theorem 2.2).

Step (χ repetition-free of length $n+1$). First record two facts. (i) The immediate subformulas of χ are repetition-free: their variables form a subset of $\mathrm{vars}(\chi)$, and no variable repeats in χ, so none repeats in a subformula. (ii) In the binary case $\chi = (\theta \circ \psi)$, $\mathrm{vars}(\theta) \cap \mathrm{vars}(\psi) = \varnothing$: a common variable would occur once in θ and once in ψ, hence at least twice in χ, contradicting repetition-freeness. Also θ, ψ have length $\leq n$.

- $\chi = \neg\theta$. By the IH there are assignments a, b with $a(\theta) = \mathrm{T}$ and $b(\theta) = \mathrm{F}$; then $a(\neg\theta) = \mathrm{F}$ and $b(\neg\theta) = \mathrm{T}$, so $\neg\theta$ takes both values.
- $\chi = (\theta \circ \psi)$, $\circ \in \{\wedge, \vee, \rightarrow, \leftrightarrow\}$. By (ii) $\mathrm{vars}(\theta) \cap \mathrm{vars}(\psi) = \varnothing$, so the Gluing Lemma (a) applies: for *any* target pair $(s, t) \in \{\mathrm{T}, \mathrm{F}\} \times \{\mathrm{T}, \mathrm{F}\}$ there is an assignment x with $x(\theta) = s$ and $x(\psi) = t$ (take u realizing $\theta = s$ and w realizing $\psi = t$, available by the IH, and glue). Now the shared feature of all five connectives' truth tables is that **each output column is non-constant — it contains at least one** T **and at least one** F:

	(T, T)	(T, F)	(F, T)	(F, F)
$\wedge$	T	F	F	F
$\vee$	T	T	T	F
$\rightarrow$	T	F	T	T
$\leftrightarrow$	T	F	F	T

  So some pair (s, t) makes $(\theta \circ \psi)$ true and some other pair makes it false; gluing realizes each, giving assignments that make χ true and false.

This closes the induction. $\square$

(c) *Exercise 2.30.* A formula built from $\neg$ and $\rightarrow$ with no repeated variable is repetition-free and uses connectives in S, so by (b) it is false under some assignment; hence it is not a tautology. *Stronger conclusion delivered by the proof:* (b) shows such a formula is also true under some assignment, so it is neither a tautology nor a contradiction — it is *contingent*; and this holds for repetition-free formulas over the *full* set $S = \{\neg, \wedge, \vee, \rightarrow, \leftrightarrow\}$, not only the $\{\neg, \rightarrow\}$ fragment. (A length-0 repetition-free formula, a single variable, is likewise contingent.)

(d) Take $\chi = (p \to p)$, which repeats p. It is a tautology, so the conclusion of (b) — false under some assignment — fails for it. The step that breaks is the binary case: here $\theta = \psi = p$, so $\mathrm{vars}(\theta) \cap \mathrm{vars}(\psi) = \{p\} \neq \varnothing$, and the disjointness on which (b) relied does not hold. Consequently the Gluing Lemma does not apply, and the pair $(s, t) = (\mathrm{T}, \mathrm{F})$ needed to make $(p \to p)$ false is *unrealizable*: it would require an assignment with $x(p) = \mathrm{T}$ and $x(p) = \mathrm{F}$ at once. Only (T, T) and (F, F) occur, and $\to$ is true on both — hence the tautology. The precise failing move is "$\mathrm{vars}(\theta) \cap \mathrm{vars}(\psi) = \varnothing$, so all four (s, t) are realizable."

(e) The freedom used in (a) — setting $x(\theta)$ and $x(\psi)$ to any pair of values at once — is available in (b) precisely because repetition-freeness makes $\mathrm{vars}(\theta)$ and $\mathrm{vars}(\psi)$ disjoint, so θ and ψ act like independent variables. In Problem 13(b) (Exercise 2.27(c)) there is no such guarantee: φ, ψ are arbitrary metavariables, and the map $w \mapsto (w(\varphi), w(\psi))$ may reach only some of the four value-pairs. Distinct metavariables are *not* distinct propositional variables: φ and ψ can share variables, or be logically related (e.g. $\psi = \neg\varphi$, as in 6(d)), or one can be pinned (e.g. ψ a tautology, as in 13(b)) — their values are computed downstream and may be correlated. When ψ is forced true, the pair always has second coordinate T, so the combination that would make φ a tautology is simply never demanded, and φ stays free. The analogy with independent variables is restored exactly when $\mathrm{vars}(\varphi) \cap \mathrm{vars}(\psi) = \varnothing$ — or, more generally, when $w \mapsto (w(\varphi), w(\psi))$ is onto $\{\mathrm{T}, \mathrm{F}\} \times \{\mathrm{T}, \mathrm{F}\}$; that surjectivity is what gluing supplies for repetition-free formulas and what is missing in general.

Three threads, one theorem. The uniqueness half of Theorem 2.2 drives every "agreement" result (1(e,g), 8, 9(b), and the determinacy in 5); the existence half is what lets you *prescribe* values — freely on the variables (2(d), 6, 7), across disjoint variable sets (16(a,b)), and even just to know an assignment exists at all (13(g)). The counterexample work (11, 12(c,f), 13(b,f), 14(c,e), 15(c,d)) is the discipline of not importing the independence of distinct variables into arbitrary metavariables — the single idea that 16(e) makes precise.